



Data Communications and Networking

Fourth Edition

Chapter 15

Connecting LANs, Backbone Networks, and Virtual LANs



15-1 CONNECTING DEVICES

In this section, we divide connecting devices into five different categories based on the layer in which they operate in a network.

Topics discussed in this section:

Passive Hubs

Active Hubs

Bridges

Two-Layer Switches

Routers

Three-Layer Switches

Gateways



Connecting Devices

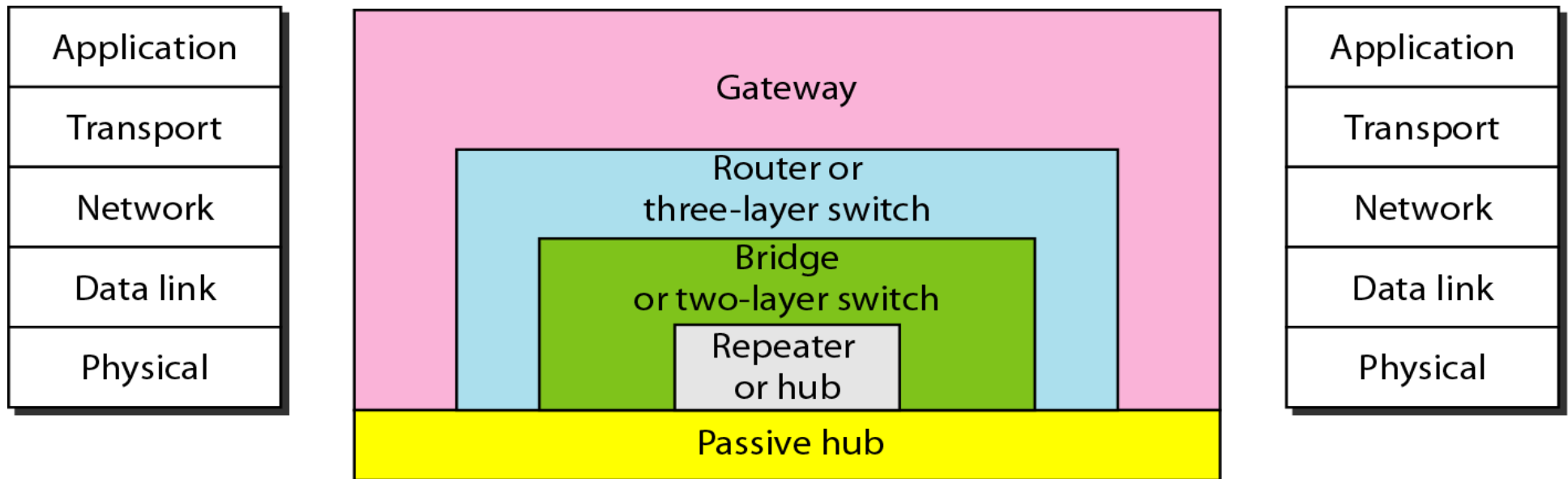
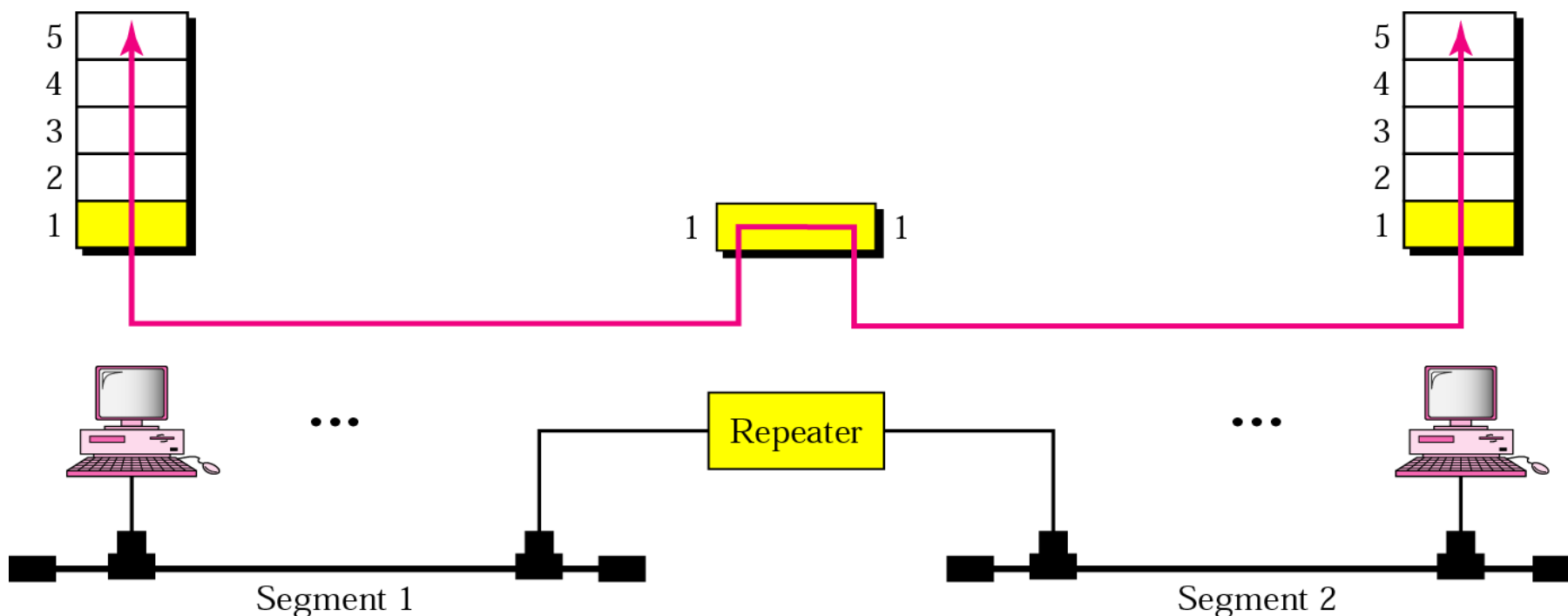


Figure 15.1 *Five categories of connecting devices*

Passive Hubs

- ❑ A passive hub is just a connector.
- ❑ It connects the wires coming from different branches.
- ❑ This type of a hub is part of the media; its location in the internet model is below the physical layer.

Repeater



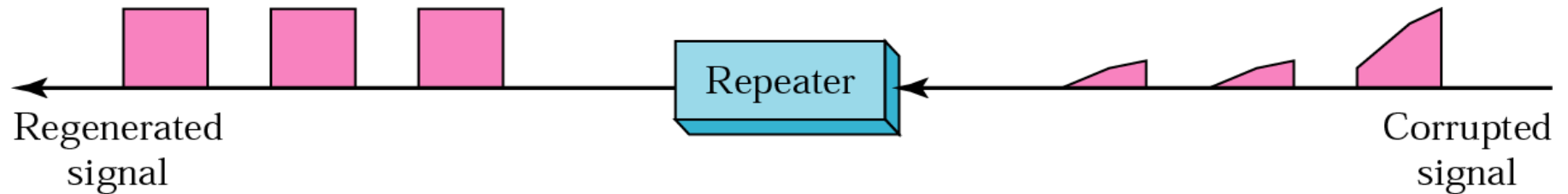
- Repeater only operates in the physical layer
- Repeater regenerates the signal, and can extend the physical length.
- Doesn't connect two LANs, connects two segments of the same LAN

Repeater

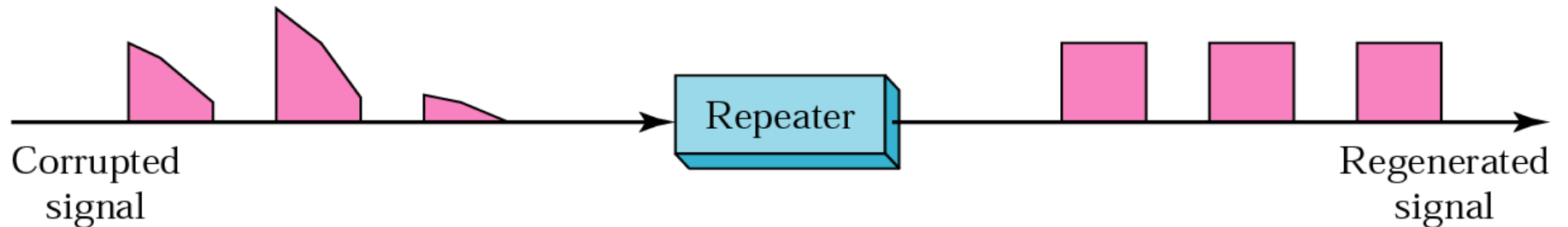
- ❑ *A repeater connects segments of a LAN.*
- ❑ *A repeater forwards every frame; it has no filtering capability*
- ❑ *A repeater is a regenerator, not an amplifier.*



Function of Repeater



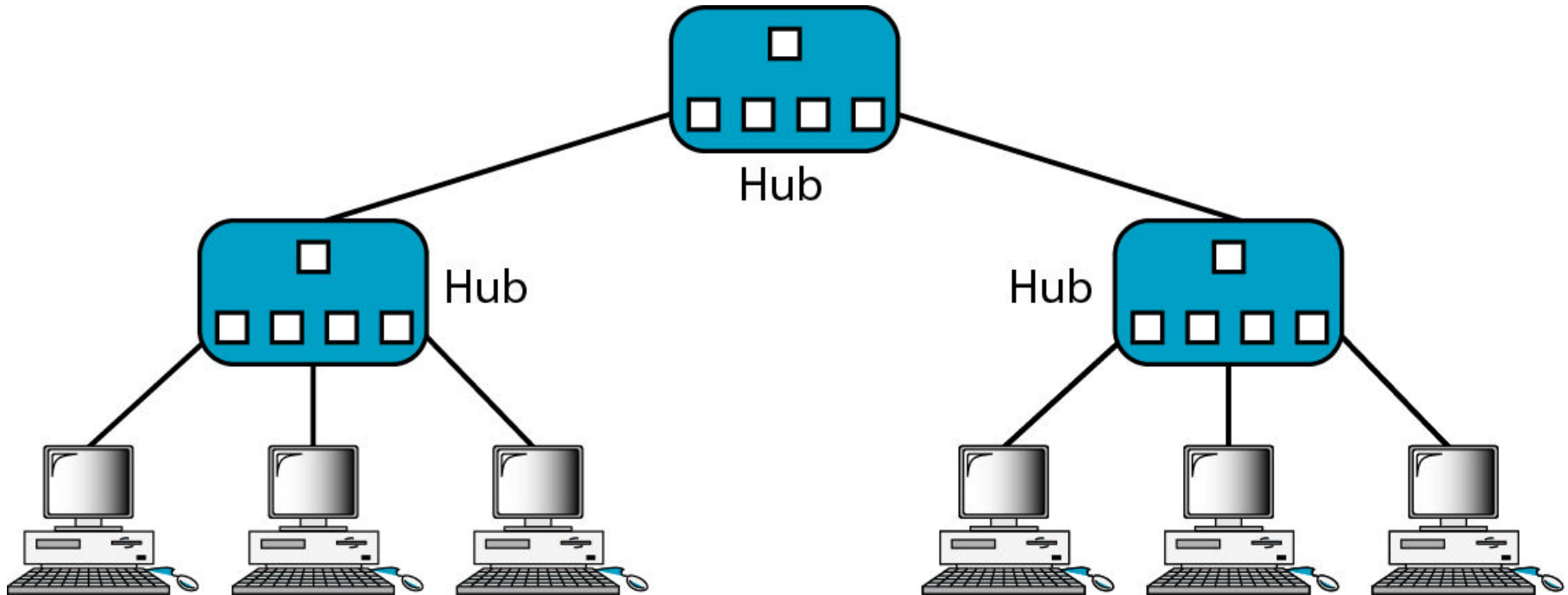
a. Right-to-left transmission.



b. Left-to-right transmission.

- Must be placed so that a signal reaches it before noise changes the meaning of its bits

Active Hubs



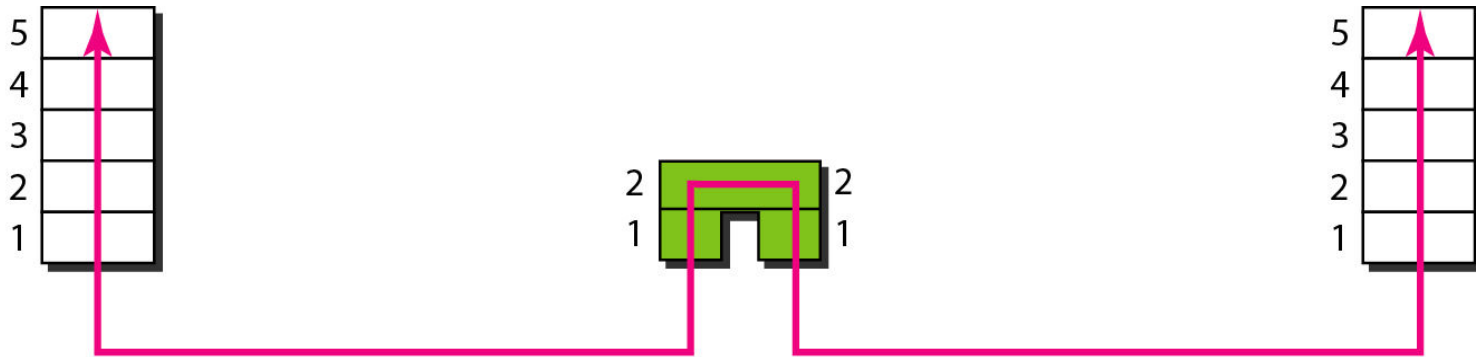
- Hub is a multiport repeater
- Creates connections between stations in a physical star topology

Bridge

- ❑ Bridge operates in both the physical and the data link layers
- ❑ As a physical layer device, it regenerates the signal
- ❑ As a data link layer device, it checks the physical (MAC) addresses
- ❑ *A bridge has a table used in filtering decisions.*



Bridge



Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	1
64:2B:13:45:61:12	2
64:2B:13:45:61:13	2

Bridge Table

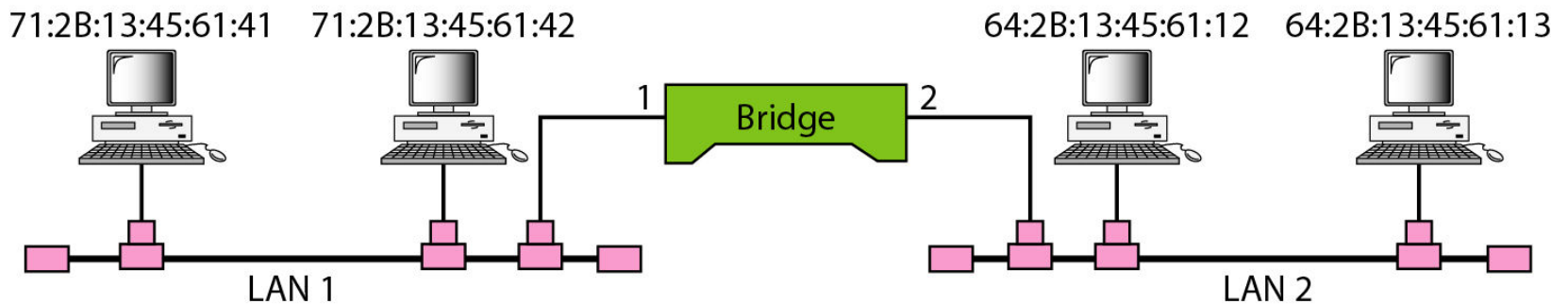


Figure 15.6 A bridge connecting two LANs

Transparent - Bridge

❑ *A bridge does not change the physical (MAC) addresses in a frame.*

❑ *Transparent Bridge*

❖ Bridge in which stations are completely unaware of the bridge's existence

❖ System equipped with transparent bridges must meet three criteria (IEEE 802.1d):

- Frames must be forwarded from one station to another
- Forwarding table is automatically made by learning from movements
- Loops must be prevented



Learning - Bridge

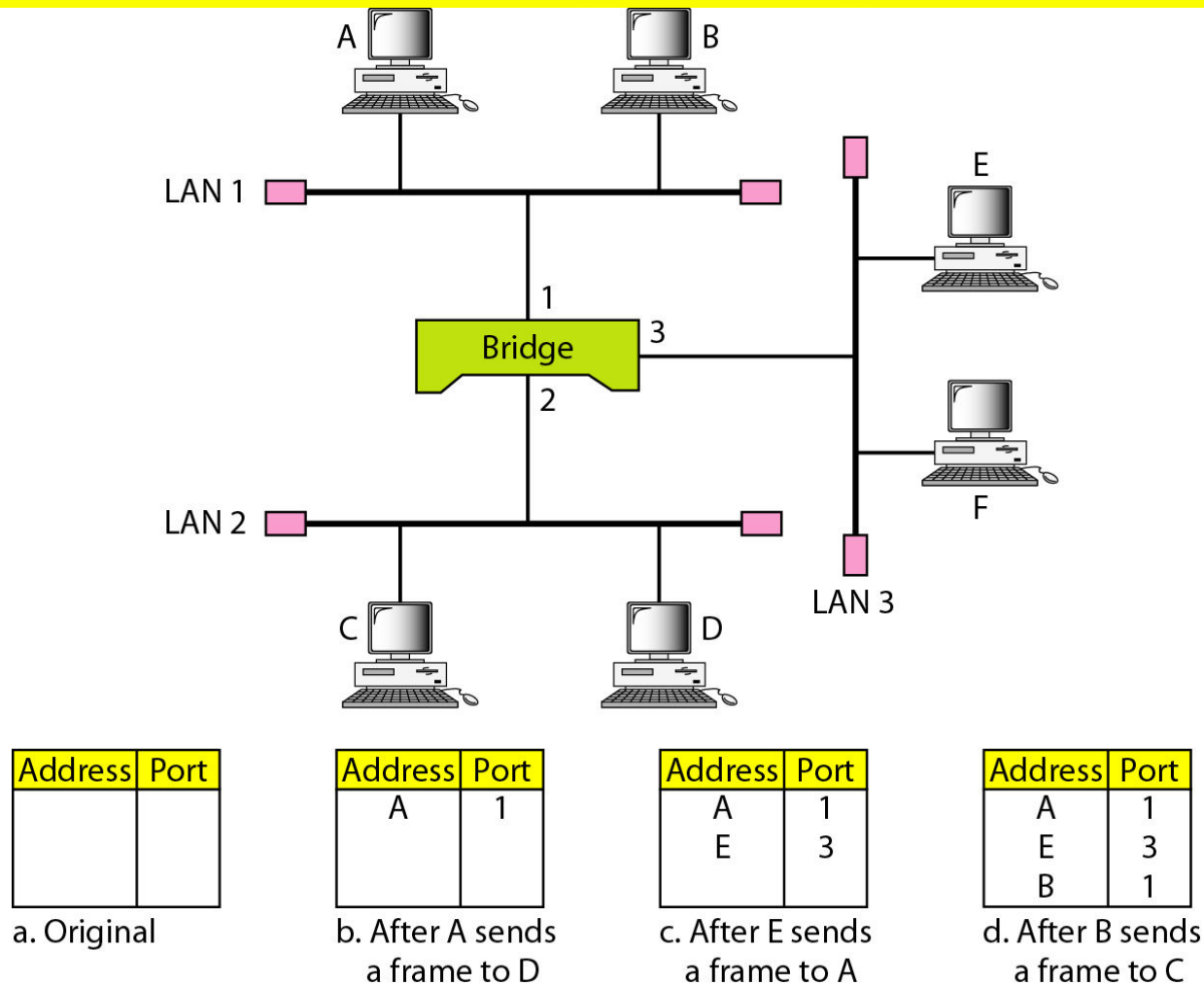


Figure 15.7 A learning bridge and the process of learning

Loop Problem

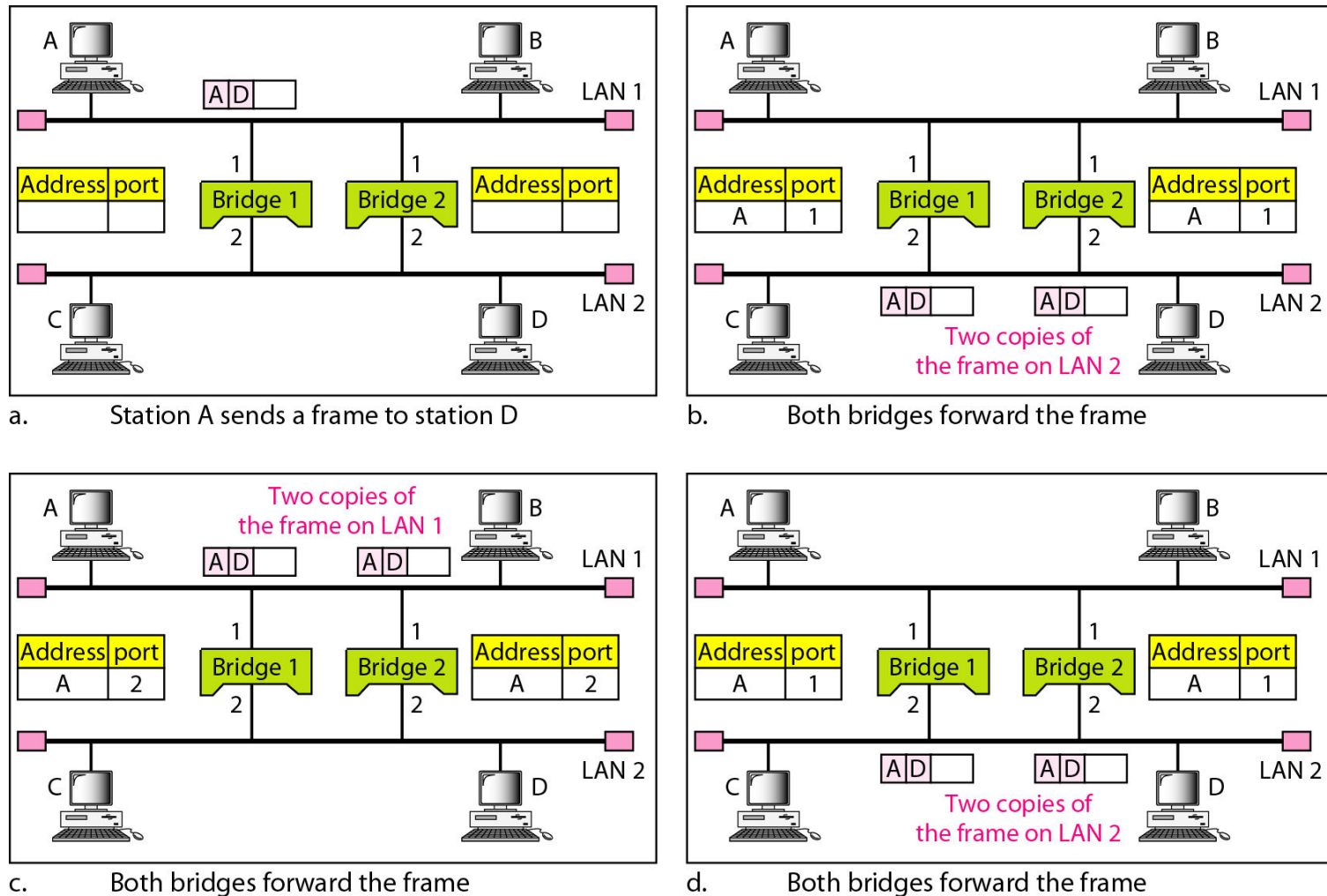


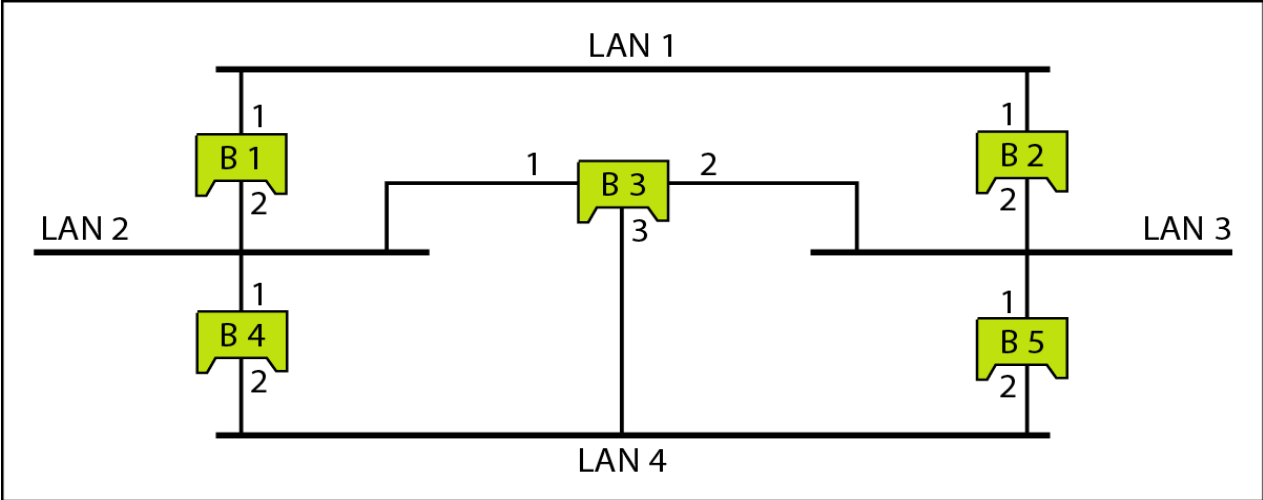
Figure 15.7 Loop problem in a learning bridge



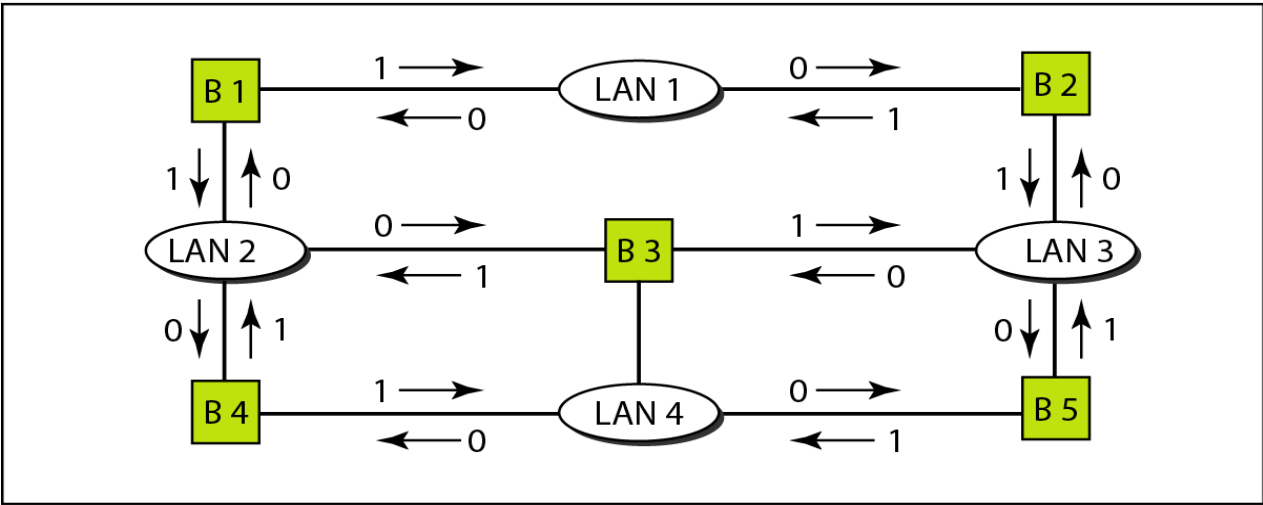
Spanning Tree

- ❑ A spanning tree is a graph in which there is no loop.
- ❑ In a bridged LAN, this means creating a topology in which each LAN can be reached from any other LAN through one path only (no loop).

Spanning Tree



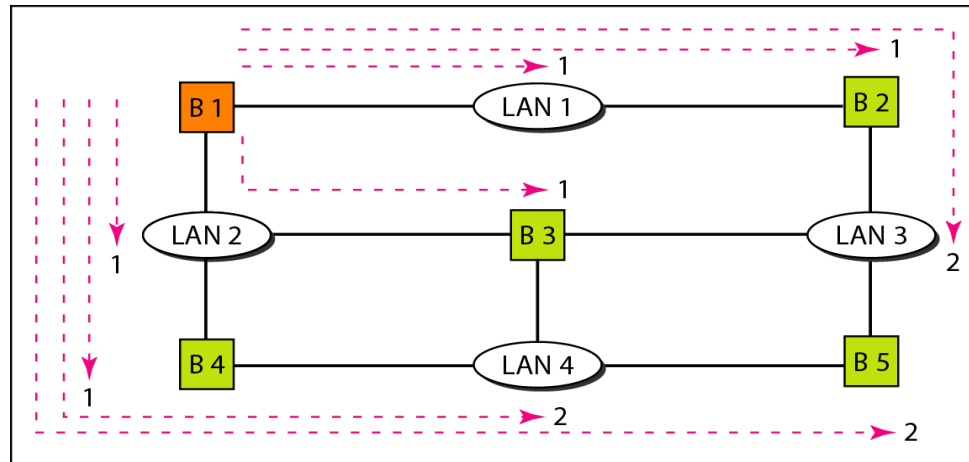
a. Actual system



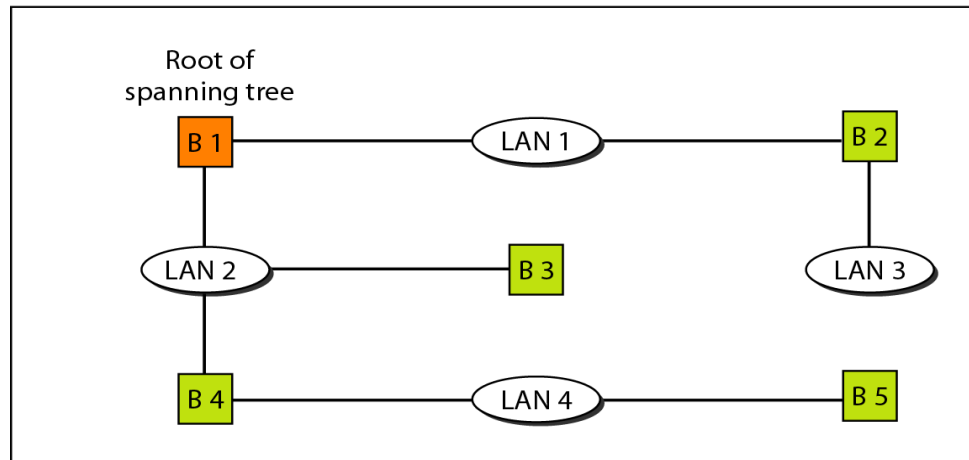
b. Graph representation with cost assigned to each arc



Spanning Tree



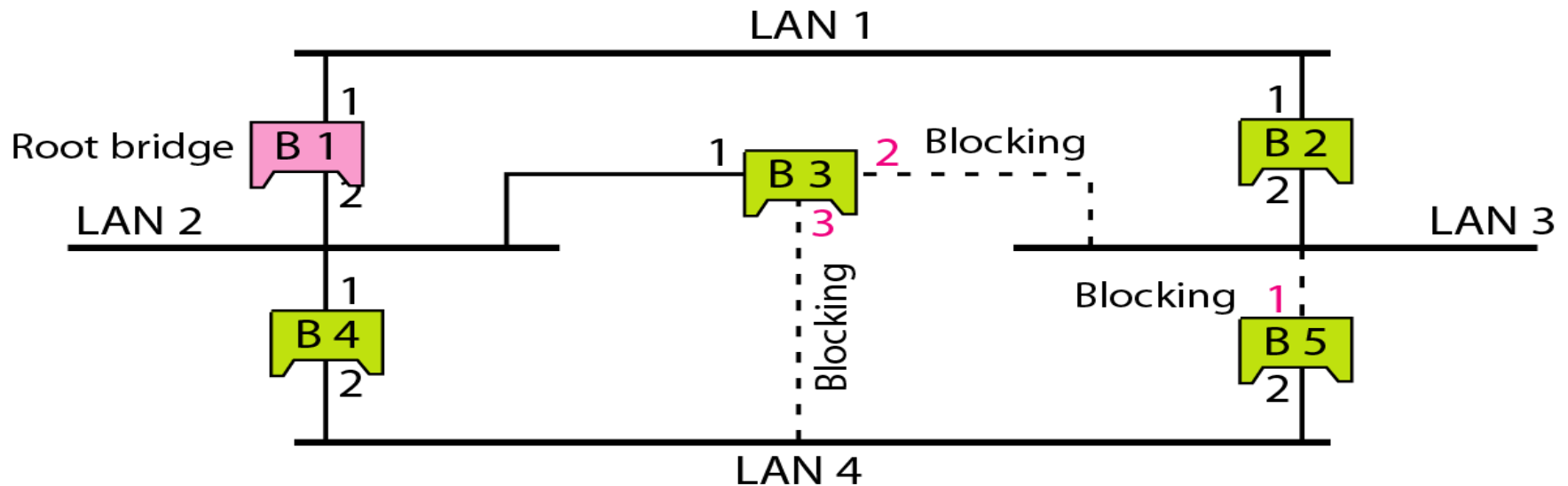
a. Shortest paths



b. Spanning tree

Figure 15.9 *Finding the shortest paths and the spanning tree in a system of bridges*

Forwarding ports and blocking ports



Ports 2 and 3 of bridge B3 are blocking ports (no frame is sent out of these ports). Port 1 of bridge B5 is also a blocking port (no frame is sent out of this port).

- Dynamic algorithm – spanning tree algorithm is done dynamically with software in the bridge using Bridge Protocol Data Unit (BPDU)

Figure 15.10 *Forwarding and blocking ports after using spanning tree algorithm*



Bridges Connecting Different LANs

❑ Bridge should be able to connect LANs using different protocols, issues to be considered:

- ❖ Frame format – Ethernet vs. wireless frame
- ❖ Max data size – too large frame must be fragmented into several frames and no protocol at the data link layer allows for fragmentation and reassembly of frames
- ❖ Data rate – each LAN has its own data rate
- ❖ Bit order – some send most significant bit first, some send least significant bit first
- ❖ Security – wireless has security measures at the data link layer, Ethernet does not
- ❖ Multimedia support – some support, some do not



Router

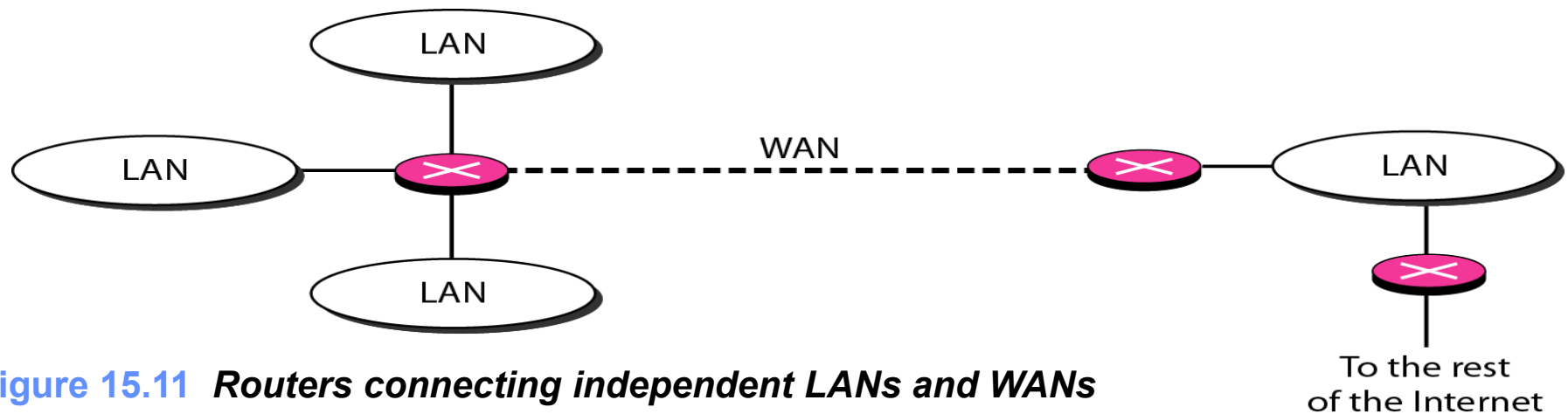


Figure 15.11 *Routers connecting independent LANs and WANs*

- ❑ A router is a three-layer device that routes packets based on their logical addresses (host-to-host addressing).
- ❑ A router normally connects LANs and WANs in the internet and has a routing table that is used for making decisions about the route.
- ❑ The routing tables are normally dynamic and are updated using routing protocols.

Gateway

- ❑ A gateway is a normally a computer that operates in all five layers of the internet or seven layers of OSI model.
- ❑ A gateway takes an application message, reads it, and interprets it.
- ❑ This means that it can be used as a connecting device between two internetworks that use different models.
 - ❖ For example, a network designed to use the OSI model can be connected to another network using the internet model.



15-2 BACKBONE NETWORKS

A backbone network allows several LANs to be connected. In a backbone network, no station is directly connected to the backbone; the stations are part of a LAN, and the backbone connects the LANs.

Topics discussed in this section:

Bus Backbone

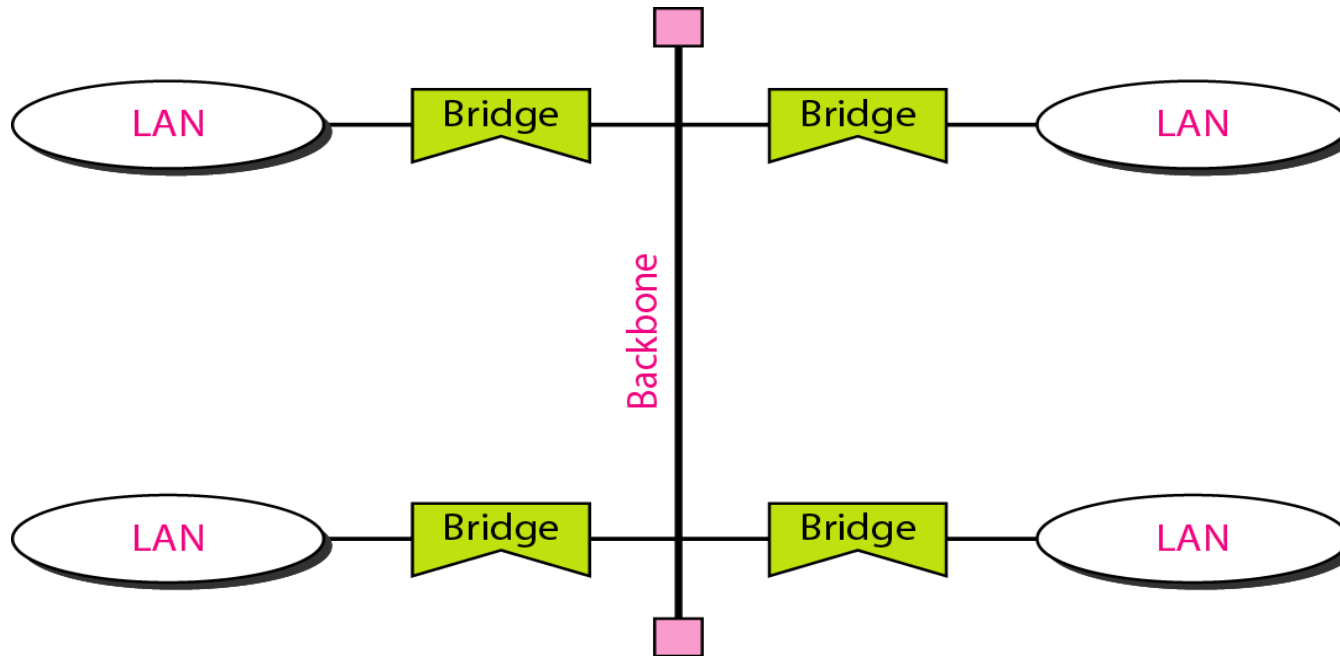
Star Backbone

Connecting Remote LANs



Bus Backbone

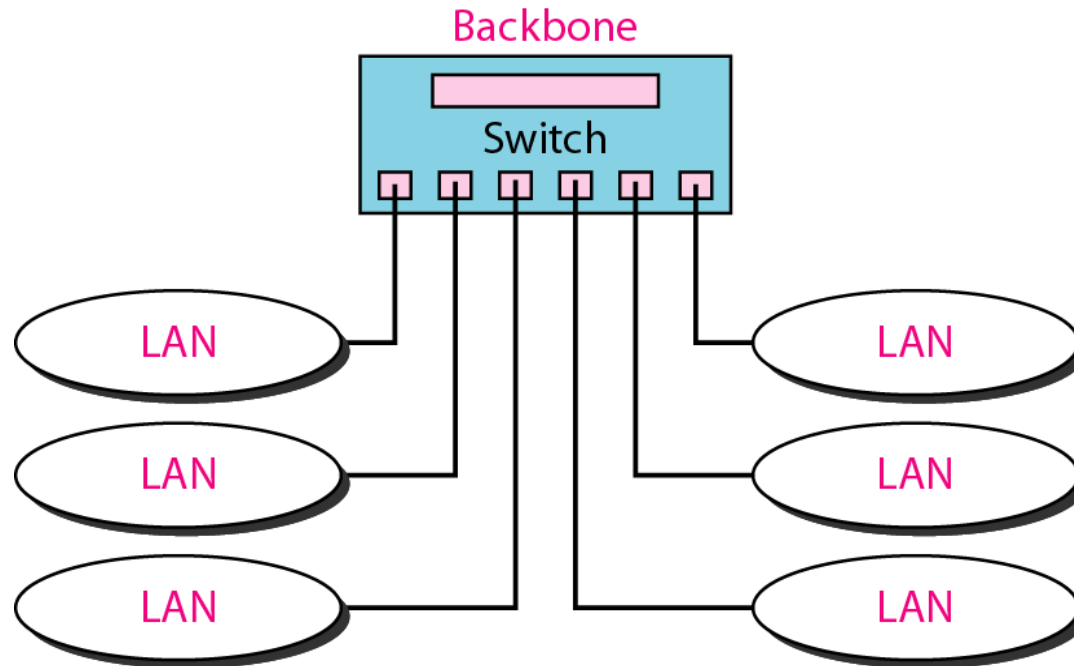
❑ *In a bus backbone, the topology of the backbone is a bus.*



- ❑ Normally used to connect different buildings in an organization
- ❑ Bridge blocks frames sent internal to the LAN
- ❑ Backbone receives frame if going from one LAN to another

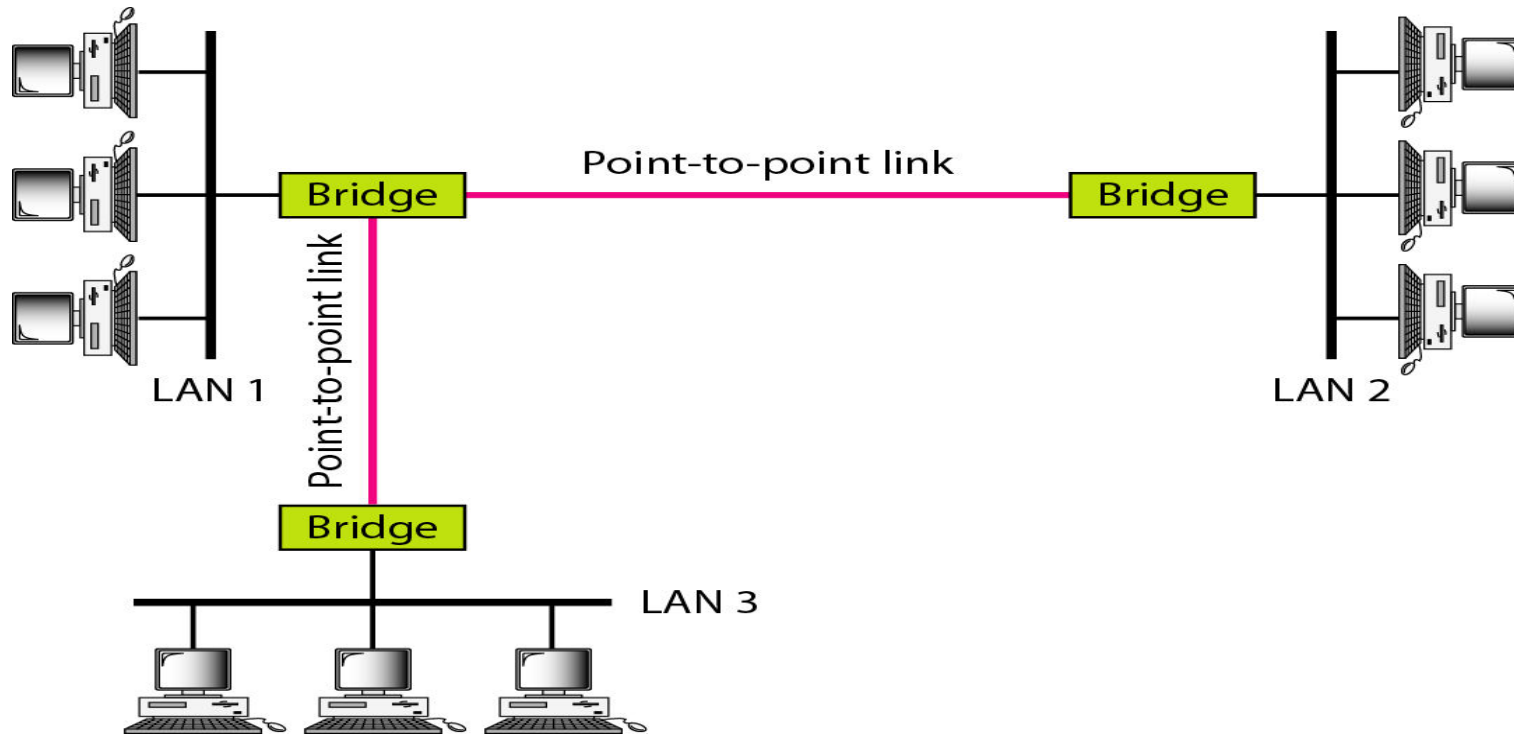
Star Backbone

- ❑ *In a star backbone, the topology of the backbone is a star; the backbone is just one switch.*



- Used as distribution backbone inside a building

Connecting remote LANs



A point-to-point link acts as a LAN in a remote backbone connected by remote bridges.

15-3 VIRTUAL LANs

We can roughly define a **virtual local area network (VLAN)** as a local area network configured by software, not by physical wiring.

Topics discussed in this section:

Membership

Configuration

Communication between Switches



Virtual LANs

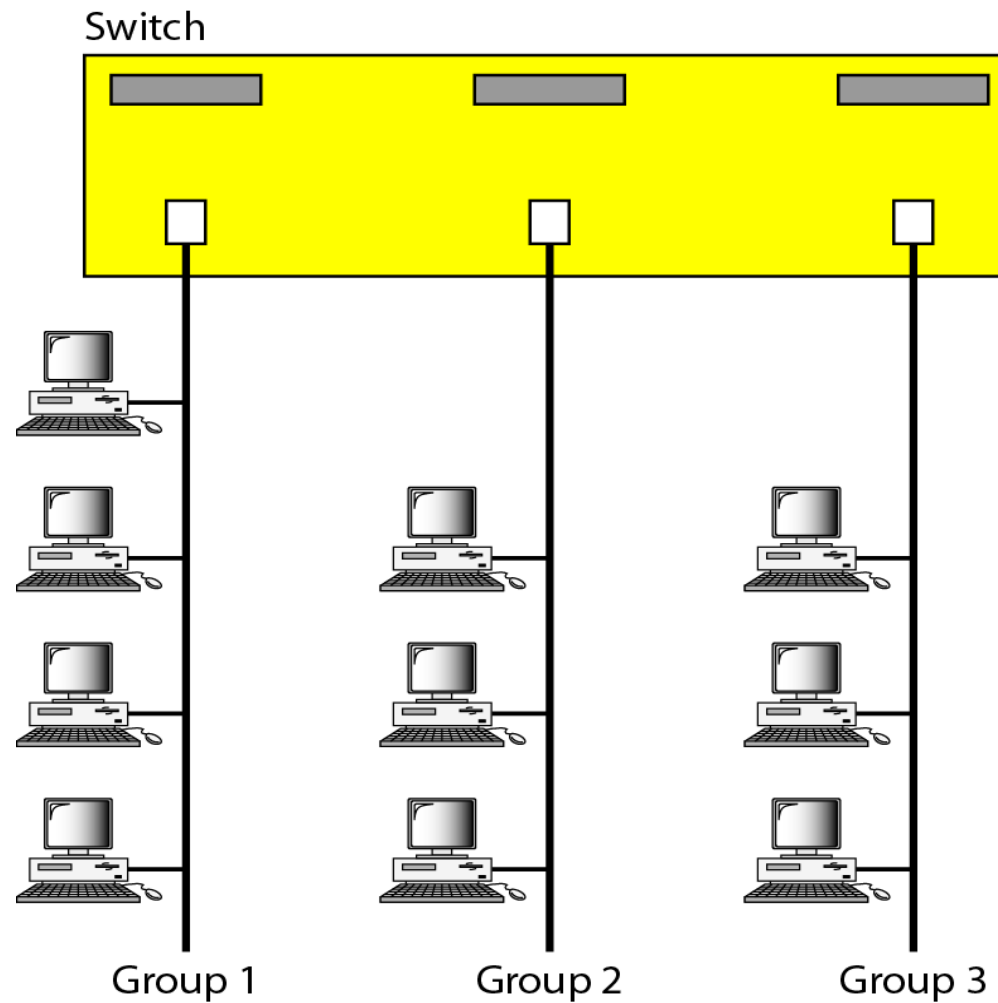


Figure 15.15 *A switch connecting three LANs*



Virtual LANs

Switch with VLAN software

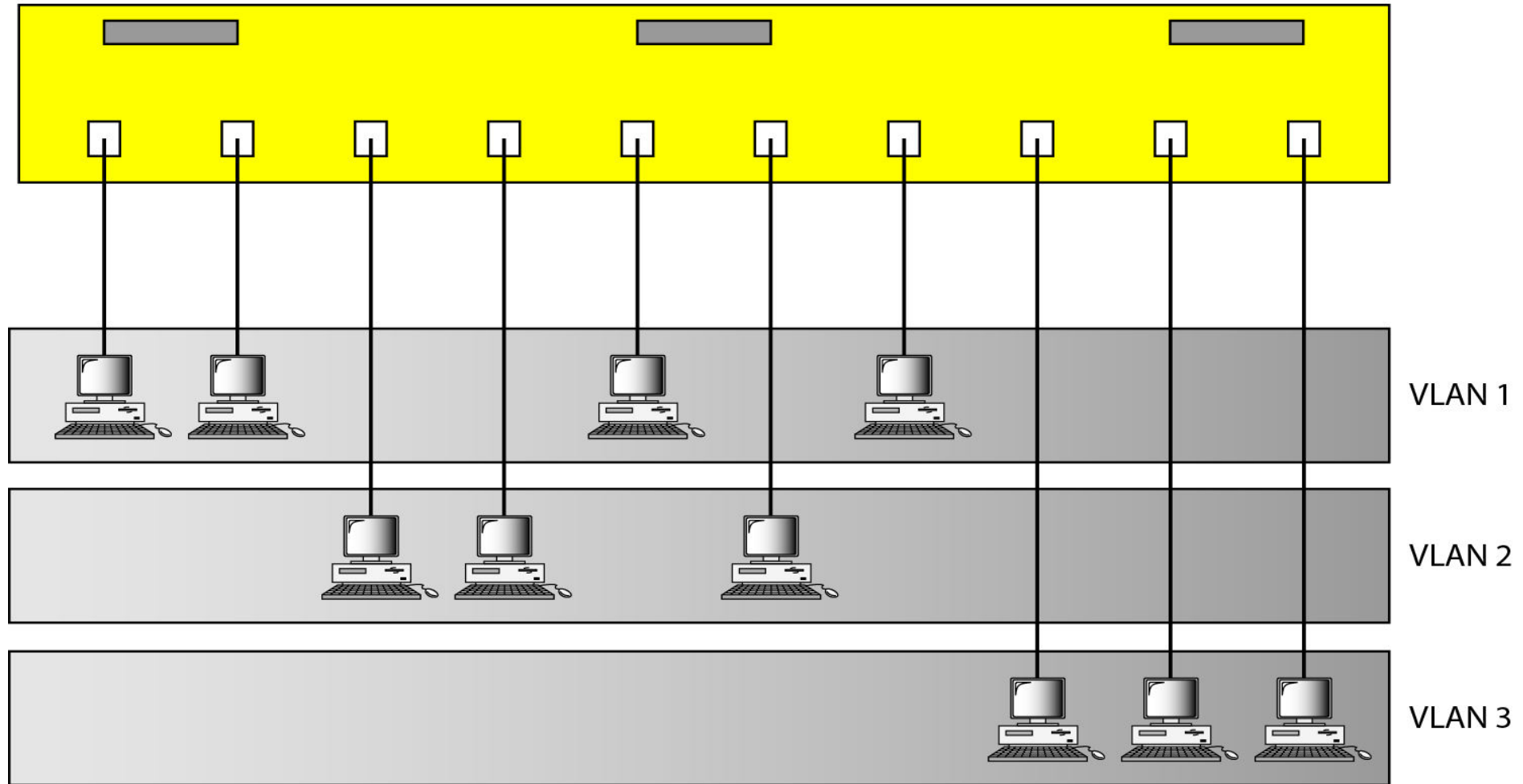
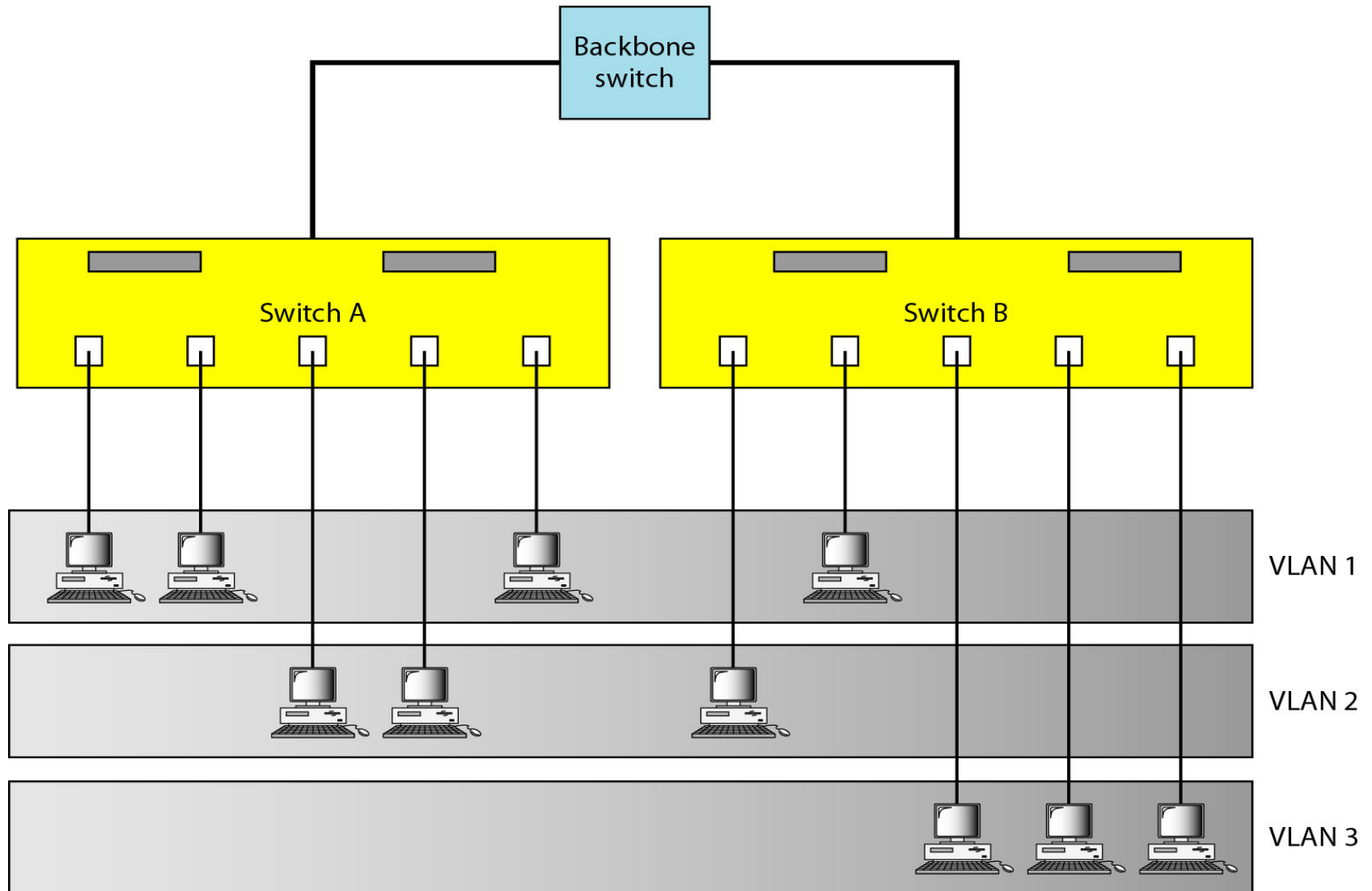


Figure 15.16 *A switch using VLAN software*



Virtual LANs



Summary

- ❑ A repeater is a connecting device that operates in the physical layer of the Internet model. A repeater regenerates a signal, connects segments of a LAN, and has no filtering capability.
- ❑ A bridge is a connecting device that operates in the physical and data link layers of the Internet model
- ❑ A transparent bridge can forward and filter frames and automatically build its forwarding table.
- ❑ A bridge can use the spanning tree algorithm to create a loopless topology.
- ❑ A backbone LAN allows several LANs to be connected.
- ❑ A backbone is usually a bus or a star.
- ❑ A virtual local area network (VLAN) is configured by software, not by physical wiring.
- ❑ Membership in a VLAN can be based on port numbers, MAC addresses, IP addresses, IP multicast addresses, or a combination of these features.
- ❑ VLANs are cost- and time-efficient, can reduce network traffic, and provide an extra measure of security.



Questions !

